

Branch Specialization Checkpoint

Conflict-Heavy Branch Routing

Branch Specialization Project

2026-05-22

Checkpoint reason: direct role conflict still did not make unlabeled routing discover causal modularity.

Prior result:

- entropy and load balancing did not reliably induce modularity;
- shrinking branches from 64 dims to 16 dims also did not help.

This checkpoint asks:

If the same query token requires incompatible predecessor-vs-successor lookups in different roles, does unlabeled routing pressure split the roles?

Task Variant

Sample a prefix:

$$[y_0, y_1, \dots, y_{15}]$$

The same query token is scored under two incompatible rules:

Role	Query	Target
Local	y_i	y_{i-1}
Induction	y_i	y_{i+1}

The task is designed so a single token-to-token lookup rule conflicts across roles.

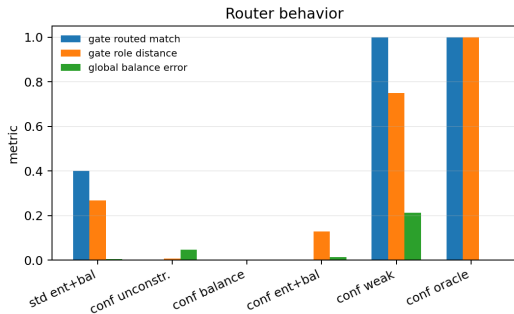
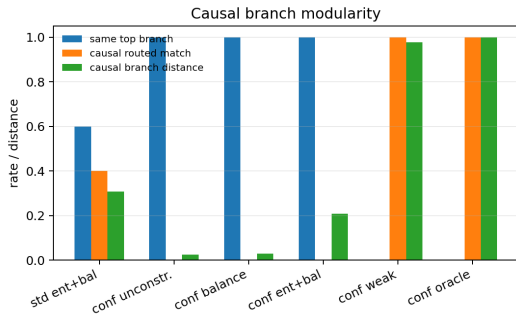
- **Same top branch:** local and induction ablations hurt most when the same branch is removed.
- **Routed role match:** local's top causal branch is branch 0 and induction's top causal branch is branch 1.
- **Branch distance:** total-variation distance between local and induction causal branch distributions.
- **Gate routed match:** router probabilities point to branch 0 for local positions and branch 1 for induction positions.

Conflict-heavy local-vs-induction task:

- task variant: `bidirectional_lookup`;
- branch attention head dims: `[64]`;
- equal local and induction loss weights;
- five seeds per condition, 1600 training steps;
- causal modularity measured by branch ablation.

Conditions: unconstrained learned token router, balance-only, entropy+balance, weak-label positive control, and oracle-route upper bound.

Main Evidence



Conflict does not rescue unlabeled modularity. Weak labels and oracle routing produce clean causal role splits.

Conflict Results

Condition	Local acc.	Ind. acc.	Same top	Routed match	Branch dist.
unconstrained	0.9992	1.0000	1.00	0.00	0.0237
balance only	0.9991	0.9994	1.00	0.00	0.0283
entropy+balance	0.9997	0.9998	1.00	0.00	0.2069
weak label	0.9986	0.9997	0.00	1.00	0.9773
oracle route	0.9994	0.9995	0.00	1.00	1.0000

All conditions solve the task, so the unlabeled failure is not a learnability failure.

Comparison to Standard Task

Condition	Routed match	Branch dist.
standard wide64 entropy+balance	0.40	0.3082
conflict wide64 entropy+balance	0.00	0.2069
standard wide64 weak label	1.00	0.8957
conflict wide64 weak label	1.00	0.9773
conflict wide64 oracle route	1.00	1.0000

Direct role conflict made the task stricter, but did not create spontaneous unlabeled causal modularity.

Supported:

- Conflict task is learnable.
- Correct routing is sufficient for clean modularity.
- Weak role-informative routing pressure is sufficient.

Not supported:

- direct predecessor-vs-successor conflict plus generic entropy/balance routing pressure discovers causal branch modularity.

Do not claim that structural modularity by itself causes functional modularity.

Stronger current claim:

Branch structure plus role-informative routing pressure can create functional modularity; branch structure plus generic unlabeled routing pressure often does not, even with direct role conflict.

Next test: anneal weak labels to determine whether the role signal is needed only early as a symmetry breaker or throughout training.

Sources and Provenance

- Report: `doc/phase3_toy_conflict_router.md`
- Model script: `scripts/toy_branch_isolation_intervention.py`
- Analysis script: `scripts/analyze_conflict_router_experiment.py`
- Analysis output: `results/phase3_toy_conflict_router_analysis/`
- Deck data snapshot: `presentations/2026-05-22-1626-conflict-router/data/`
- Representative command: `python3 -u scripts/toy_branch_isolation_intervention.py --task-variant bidirectional_lookup`